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5th Sem/ Automobile
Subject : Auto Electrical Systems & Electric Vehicles

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The positive plate of lead acid battery has

- a) Lead peroxide (PbO_2)
- b) Spongy Lead (Pb)
- c) Lead sulphate (PbSO_4)
- d) Sulphuric acid (H_2SO_4)

Q.2 What is maintenance free battery?

- a) A battery having lead antimony plate grid
- b) A battery having lead calcium plate grid
- c) A battery does not contain acid
- d) A battery does not contain water

Q.3 At the start of the engine, the charging voltage is

- a) Lower
- b) Higher
- c) Same
- d) Zero

- Q.4 The brake warning light warns the driver of ____
- Water in the master cylinder
 - Air in the hydraulic system
 - Failure of the primary or secondary circuit of hydraulic system
 - Power brake failure
- Q.5 The parking brake generally acts on ____
- Front wheels
 - Rear wheel
 - Propeller shaft
 - Front & rear both
- Q.6 The cause of Defective wiper may be
- Wire disconnected
 - Loose connection
 - Blown
 - All of above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Dynamo generates ____ current.
- Q.8 Electrolyte used in lead acid battery is ____.
- Q.9 Function of Fog light used in automobiles is ____.
- Q.10 What is the purpose of Indicator in automobiles?
- Q.11 What is the main function of odometer in automobile?
- Q.12 Write the function of windscreen wiper in automobile?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write various specifications of Battery.
- Q.14 Write charging and discharging equations of Lead Acid Battery.
- Q.15 Explain the concept of low maintenance battery.
- Q.16 Describe the working of dynamo with diagram.
- Q.17 What are the advantages of alternator over dynamo?
- Q.18 Write any four cable colour codes.
- Q.19 Describe the significance of wiring harness.
- Q.20 Describe the working of defogger.
- Q.21 Describe advance driver Assistant system.
- Q.22 Explain Hybrid vehicles with diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe the various methods of testing of battery. Explain any two in detail.
- Q.24 Explain with neat sketch explain wiring diagram of a modern car. What are various circuits in wiring diagram?
- Q.25 Draw the diagram of fuel cell and explain principal, working and types of fuel cell.